Mengyu Chang

Occupation: Postdoctororal Fellow
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Hometown: Liaoning Province, China
Address: Department of Radiation Oncology, The University of Texas MD Anderson Cancer Center, Houston, TX, 77030, USA.

Education

Postdoctor	2023.01 – now	Department of Radiation Oncology, The University of Texas MD Anderson Cancer Center, USA Supervisors: Dr. Wen Jiang
Postdoctor	2022.02 – 2022.12	School of Chemistry, Chemical Engineering and Biotechnology, Nanyang Technological University, Singapore Supervisors: Dr. Yanli Zhao
Research Assistant	2021.07 - 2021.12	Changchun Institute of Applied Chemistry, Chinese Academy of Sciences, China Supervisors: Dr. Jun Lin
Ph.D.	2016.09 - 2021.06	Changchun Institute of Applied Chemistry, Chinese Academy of Sciences, China Supervisors: Dr. Jun Lin
Bachelor	2012.09 - 2016.06	College of Chemistry and Chemical Engineering, Hunan Normal Univesity, China Supervisor: Dr. Shixun Lian

Professional Skill

1. **Nanotechnologies**
   Construction and synthesis of multifunctional nanomaterials including single and composite nanostructures (rare earth compounds, transition metal chalcogenides, single atom nanozymes, metal alloy, metallic oxide), as well as surface modification of nanomaterials by noble metal deposition, in situ mineralization and macromolecule coating. Using multifunctional nanomaterials for multimodal imaging (photothermal/photoacoustic/ computed tomography imaging) mediated synergistic anti-tumor therapy (photothermal/ photodynamic/chemodynamic/starvation/ions interference/ferroptotic therapy and immunotherapy)
2. **Basic Molecular Biology Skills**
   Cell culture, immunofluorescence, enzyme-linked immunosorbent assay (E-LISA), and histological section

3. **Basic Animal Experiment Skills**
   Intravenous (i.v.), intraperitoneal (i.p.), and subcutaneous (s.c.) injection, small animal anesthesia/blood collection/anatomy, and single cell suspension obtained from tissue (including spleen, lymph gland and tumor)
4. **Materials Characterization Instruments**
   X-ray diffractometer (XRD), scanning electron microscope (SEM), transmission electron microscope (TEM), UV-visible diffuse reflectance and absorption spectra, dynamic light scattering (DLS), Zeta potential, MSOT inVision 128 photoacoustic imaging and flow cytometry.

Honors&Awards

2022	Prof Lee Soo Ying Young Chemist Award, Gold Award, Singapore National Institute of Chemistry
2021	Outstanding Graduate of Anhui Province
2021	Outstanding Graduate of University of Science and Technology of China
2021	President Award of Chinese Academy of Sciences
2021	Excellent Doctoral Dissertation of University of Science and Technology of China
2020	National Scholarship
2016-2020	School-level Scholarship
2018	Excellent Student Cadre
2017	Merit Student

Publications

([#] Co-first author, * Corresponding author)

1. **Mengyu Chang,** Zhiyao Hou*, Man Wang, Ding Wen, Chunxia Li, Yuhui Liu, Yanli Zhao*, and Jun Lin*, Cu Single Atom Nanozyme Based High-Efficiency Mild Photothermal Therapy through Cellular Metabolic Regulation, ***Angew. Chem. Int. Ed.*** 2022, 61, e20220924. DOI: 10.1002/anie.202209245
2. **Mengyu Chang**, Zhiyao Hou*, Man Wang, Chunzheng Yang, Ruifeng Wang, Fang Li, Donglian Liu, Tieli Peng, Chunxia Li*, and Jun Lin*, Single-Atom Pd Nanozyme for Ferroptosis-Boosted Mild-Temperature Photothermal Therapy, ***Angew. Chem. Int. Ed.*** 2021, 60, 12971–12979. DOI: 10.1002/anie.202101924
3. **Mengyu Chang**, Zhiyao Hou*, Dayong Jin*, Jiajia Zhou, Man Wang, Meifang Wang, Mengmeng Shu, Binbin Ding, Chunxia Li, and Jun Lin*, Colorectal Tumor Microenvironment-Activated Bio-Decomposable and Metabolizable Cu_2O@$CaCO_3$ Nanocomposites for Synergistic Oncotherapy, ***Adv. Mater.***, 2020, 32, 2004647. DOI: 10.1002/adma.202004647
4. **Mengyu Chang**, Man Wang, Meifang Wang, Mengmeng Shu, Binbin Ding, Chunxia Li, Maolin Pang, Shuzhong Cui, Zhiyao Hou*, and Jun Lin*, A Multifunctional Cascade Bioreactor Based on Hollow-Structured Cu_2MoS_4 for Synergetic Cancer Chemo-Dynamic Therapy/Starvation Therapy/Phototherapy/Immunotherapy with Remarkably Enhanced Efficacy, ***Adv. Mater.***, 2019, 31, 1905271. DOI: 10.1002/adma.201905271
5. **Mengyu Chang**, Zhiyao Hou*, Man Wang, Chunxia Li, and Jun Lin*, Recent Advances in Hyperthermia Therapy-Based Synergistic Immunotherapy, ***Adv. Mater.***, 2020, 33, 2004788. DOI: 10.1002/adma.202004788

6. **Mengyu Chang**, Man Wang, Yuhui Liu, Min Liu, Abdulaziz A. Al Kheraif, Ping'an Ma,* Yanli Zhao,* and Jun Lin*, Dendritic Plasmonic CuPt Alloys for Closed-Loop Multimode Cancer Therapy with Remarkably Enhanced Efficacy, ***Small***, 2022, 2206423. DOI: 10.1002/smll.202206423
7. **Mengyu Chang**, Zhiyao Hou*, Man Wang, Meifang Wang, Peipei Dang, Jianhua Liu, Mengmeng Shu, Binbin Ding, Abdulaziz A. Al Kheraif, Chunxia Li*, and Jun Lin*, Cu_2MoS_4/Au Heterostructures with Enhanced Catalase-Like Activity and Photoconversion Efficiency for Primary/Metastatic Tumors Eradication by Phototherapy-Induced Immunotherapy, ***Small***, 2020, 16, 1907146. DOI: 10.1002/smll.201907146
8. **Mengyu Chang**, Meifang Wang, Mengmeng Shu, Yajie Zhao, Binbin Ding, Shanshan Huang, Zhiyao Hou*, Gang Han, and Jun Lin*, Enhanced Photoconversion Performance of $NdVO_4$/Au Nanocrystals for Photothermal/Photoacoustic Imaging Guided and Near Infrared Light-Triggered Anticancer Phototherapy, ***Acta Biomater.***, 99 (2019) 295–306. DOI: 10.1016/j.actbio.2019.08.026
9. **Mengyu Chang**, Meifang Wang, Yeqing Chen, Mengmeng Shu, Yajie Zhao, Binbin Ding, Zhiyao Hou*, and Jun Lin*, Self-assembled $CeVO_4$/Ag Nanohybrid as Photoconversion Agents with Enhanced Solar-Driven Photocatalysis and NIR-Responsive Photothermal/ Photodynamic Synergistic Therapy Performance, ***Nanoscale***, 2019, 11, 10129. DOI: 10.1039/c9nr02412c
10. **Mengyu Chang**, Zhiyao Hou*, Man Wang, Chunxia Li, Abdulaziz A. Al Kheraif, and Jun Lin*, Tumor Microenvironment Responsive Single-Atom Nanozymes for Enhanced Antitumor Therapy, ***Chem. Eur. J.***, 2022, 28, e202104081. DOI: 10.1002/chem.202104081
11. **Mengyu Chang**, Man Wang, Zhiyao Hou*, and Jun Lin*, Problems and Solutions of Nanomaterials in Antitumor Photothermal Therapy, ***Chinese J. Lumin.***, 2022. (in Chinese) DOI: 10.37188/CJL.20220118
12. Man Wang#, **Mengyu Chang#**, Chunxia Li*, Qing Chen, Zhiyao Hou*, Bengang Xing, and Jun Lin, Tumor-Microenvironment-Activated Reactive Oxygen Species Amplifier for Enzymatic Cascade Cancer Starvation/Chemodynamic /Immunotherapy, ***Adv. Mater.***, 2022, 34, 2106010. DOI: 10.1002/adma.202106010
13. Man Wang#, **Mengyu Chang#**, Qing Chen, Dongmei Wang, Chunxia Li*, Zhiyao Hou*, Jun Lin*, Dayong Jin, and Bengang Xing, Au_2Pt-PEG-Ce6 Nanoformulation with Dual Nanozyme Activities for Synergistic Chemodynamic Therapy/Phototherapy, ***Biomaterials***, 252 (2020) 120093. DOI: 10.1016/j.biomaterials.2020.120093
14. Yuhui Liu, Yaning Lu, Shuang Zhang, Xiaoyan Li, Zhibin Zhang, Liya Ge, **Mengyu Chang***, Yunhai Liu*, Grzegorz Lisak, and Sheng Deng*, Amphiphilic ligand in situ assembly of uranyl active sites and selective interactions of molybdenum disulfide, ***J. Hazard. Mater.***, Volume 442, 2023, 130089, ISSN 0304-3894. DOI: 10.1016/j.jhazmat.2022.130089
15. Shuang Zhang, Fan Yang, Xiaohui Cao, Yong Tang, Taiqi Yin, Tao Bo, Yunhai Liu, Grzegorz Lisak, Naoki Kano, Bing Na*, **Mengyu Chang***, Yuhui Liu*, Enhanced uranium separation by charge enabling γ-MnO2 with oxygen vacancies, ***J. Hazard. Mater.***, Volume 459, 2023, 132112, ISSN 0304-3894. DOI: 10.1016/j.jhazmat.2023.132112.
16. Chunzheng Yang, Man Wang, **Mengyu Chang**, Meng Yuan, Wenying Zhang, Jia Tan, Binbin Ding, Ping'an Ma*, and Jun Lin*, Heterostructural Nanoadjuvant CuSe/$CoSe_2$ for Potentiating Ferroptosis and Photoimmunotherapy through Intratumoral Blocked Lactate Efflux, ***J. Am. Chem. Soc.***, 2023, 145, 13, 7205–7217. DOI: 10.1021/jacs.2c12772

17. Binbin Ding, Pan Zheng, Fan Jiang, Yajie Zhao, Meifang Wang, **Mengyu Chang**, Ping'an Ma*, and Jun Lin*, MnO_x Nanospikes as Nanoadjuvants and Immunogenic Cell Death Drugs with Enhanced Antitumor Immunity and Antimetastatic Effect, ***Angew. Chem. Int. Ed.***, 2020, 59, 1–5. DOI: 10.1002/anie.202005111
18. Chunzheng Yang, Jiashi Zhang, **Mengyu Chang**, Jia Tan, Meng Yuan, Yulong Bian, Bin Liu, Zhendong Liu, Meifang Wang, Binbin Ding, Ping'an Ma*, and Jun Lin*, NIR-Activatable Heterostructured Nanoadjuvant CoP/NiCoP Executing Lactate Metabolism Interventions for Boosted Photocatalytic Hydrogen Therapy and Photoimmunotherapy. ***Adv. Mater.***, 2023, 2308774. DOI: 10.1002/adma.202308774
19. Man Wang, **Mengyu Chang**, Pan Zheng, Qianqian Sun, Guangqiang Wang, Jun Lin*, Chunxia Li*, A Noble AuPtAg-GOx Nanozyme for Synergistic Tumor Immunotherapy Induced by Starvation Therapy-Augmented Mild Photothermal Therapy. ***Adv. Sci.***, 2022, 9, 2202332. DOI: 10.1002/advs.202202332
20. Man Wang, Chunzheng Yang, **Mengyu Chang**, Yulin Xie, Guoqing Zhu, Yanrong Qian, Pan Zheng, Qianqian Sun, Jun Lin*, and Chunxia Li*, Single-atom nanozymes based nanobee vehicle for autophagy inhibition-enhanced synergistic cancer therapy, ***Nano Today***, 52 (2023) 101981. DOI: 10.1016/j.nantod.2023.101981.
21. Chunzheng Yang, **Mengyu Chang**, Meng Yuan, Fan Jiang, Binbin Ding, Yajie Zhao, Peipei Dang, Ziyong Cheng, Abdulaziz A. Al Kheraif, Ping'an Ma*, and Jun Lin*, NIR-Triggered Multi-Mode Antitumor Therapy Based on Bi_2Se_3/Au Heterostructure with Enhanced Efficacy, ***Small***, 2021, 17, 2100961. DOI: 10.1002/smll.202100961
22. Meifang Wang, Zhiyao Hou*, Sainan Liu, Shuang Liang, Binbin Ding, Yajie Zhao, **Mengyu Chang**, Gang Han, Abdulaziz A. Al Kheraif, J. Lin*, A Multifunctional Nanovaccine based on L-Arginine-Loaded Black Mesoporous Titania: Ultrasound-Triggered Synergistic Cancer Sonodynamic Therapy/Gas Therapy/Immunotherapy with Remarkably Enhanced Efficacy. ***Small*** 2021, 17, 2005728. DOI: 10.1002/smll.202005728
23. Zhiyao Hou, Kerong Deng, Meifang Wang, Yihan Liu, **Mengyu Chang**, Shanshan Huang, Chunxia Li*, Yi Wei, Ziyong Cheng, Gang Han, Abdulaziz A. Al Kheraif, and Jun Lin* Hydrogenated Titanium Oxide Decorated Upconversion Nanoparticles: Facile Laser Modified Synthesis and 808 nm NIR Light Triggered Phototherapy, ***Chem. Mater.***, 2019, 31, 774-784. DOI：10.1021/acs.chemmater.8b03762
24. Meifang Wang, Yajie Zhao, **Mengyu Chang**, Binbin Ding, Zhiyao Hou*, and Jun Lin, Azo Initiator Loaded Black Mesoporous Titania with Multiple Optical Energy Conversion for Synergetic Photo-Thermal-Dynamic Therapy, ***ACS Appl. Mater. Interfaces***, 2019, 11, 47730-47738. DOI: 10.1021/acsami.9b17375
25. Yanshu Shi#, Sainan Liu#, Ying Liu, Chunqiang Sun, **Mengyu Chang**, Xueyan Zhao, Chunling Hu, and Maolin Pang*, Facile Fabrication of Nanoscale Porphyrinic Covalent Organic Polymers for Combined Photodynamic and Photothermal Cancer Therapy, ***ACS Appl. Mater. Interfaces***, 2019, 11, 12321-12326. DOI: 10.1021/acsami.9b00361
26. Yuhui Liu, Jiayin Zhao, Ke Zhao, Shuang Zhang, Rongteng Tian, **Mengyu Chang**, Yingcai Wang, Xiaoyan Li, Liya Ge, Grzegorz Lisak, MoS_2/Au heterojunction nanostructures for electrochemical detection of trace Hg(II) in wastewater samples, ***J. Environ. Chem. Eng.***, Volume 11, Issue 6, 2023, 111563, ISSN 2213-3437, DOI: 10.1016/j.jece.2023.111563.

Google Scholar: https://scholar.google.com/citations?user=WaHuZvMAAAAJ&hl=zh-CN&oi=ao